

Learning Report – Graduation Internship

Quantum Imaging with Undetected Photons: Enhanced Image processing & User Interface

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1 Reflection on used methodology

Looking back at how I managed my workflow, choosing the Scrum framework was definitely the right call for building this demonstrator. In the beginning, the assignment was kept pretty open. The main requirements were just that the system had to be programmed entirely in Python and that it needed to calculate the necessary images. Because it was so broad, I actually felt pretty lost at first and didn't really know what to expect.

This is where the weekly sprint meetings really helped me. Discussing the project every week was especially useful during the early research phase when I was trying to wrap my head around the physical principles. These discussions didn't just give me a solid understanding of what QIUP actually is, but they also gave me clear direction on how to start developing the application.

Breaking the workload down into weekly sprints allowed me to build the system step by step. I started with an initial design and tackled the project in small, manageable tasks until I reached the finalized GUI. This approach was really important because the setup depends so heavily on the sensitive nature of quantum interference. As I learned more about how the hardware actually had to behave in practice, the technical requirements needed to be adjusted. Instead of being stuck with an initial plan that no longer made sense, I could evaluate bottlenecks and adjust my approach each week.

A great example of this was the image enhancement feature. At the start of the project, enhancing the produced images was completely out of scope and hadn't even been considered. However, because Scrum allowed me to work iteratively and explore new possibilities, I was able to dive into this later on. Adding these enhancements ended up playing a big part in improving the system's capabilities and the overall user experience.

Overall, working this way taught me a valuable lesson in adaptability. Scrum gave me the flexibility to explore ideas and adjust to hardware challenges as they came up, while still keeping the final goal in mind.

2 Self-evaluation on X competences

2.1 Analyse

When I first started, the assignment was incredibly broad: build a Python setup to calculate images for QIUP. Before I could even think about code, I had to figure out what was actually happening physically. I spent a lot of time analyzing advanced research papers on spontaneous parametric down-conversion (SPDC) and nonlinear interferometry just to understand the baseline requirements. Once I grasped the physics, I had to evaluate my software approach. For example, I quickly realized that running a standard pixel-wise FFT was a massive computational bottleneck that would ruin the frame rate. I had to weigh my options and ultimately integrated the FFTW library to hit our real-time processing requirements.

2.2 Design

Because the project was so open-ended, my software architecture had to evolve organically. I couldn't just design the whole system up front. Instead, I used my weekly sprints to iteratively design a Python-based control system that could handle raw interferometric data and interface with the Thorlabs hardware. Working this way meant I could easily pivot when needed. A great example of this was when I designed the image enhancement workflows, it wasn't in the original scope, but the flexible design allowed me to integrate it smoothly, drastically improving the final user experience and the quality of the visibility maps.

2.3 Realize

Actually building and delivering the application was a huge learning curve. I took the theoretical concepts and turned them into a working demonstrator by building the image processing pipeline from scratch using OpenCV and FFTW. The hardest part to realize was the hardware integration. The Thorlabs C-based SDK was mostly undocumented, so I had to essentially reverse-engineer it to get it talking to my Python architecture. Figuring out how to handle the .NET API between C and Python, allowed me to successfully deliver a system that extracts visibility, contrast and phase information from the interference fringes.

2.4 Control

To make sure the software actually functioned optimally in a live lab setting, I had to implement some strict performance controls. Calculating phase-shifted images on the fly is computationally heavy, and I needed to ensure the system wouldn't crash or drop frames during live demonstrations. By optimizing the FFT processing and stabilizing the communication loop with the Thorlabs piezoelectric controllers, I made sure the setup was robust. I wanted to leave behind a maintainable, reliable tool that the research group could actually keep using for their experiments long after my internship ended.

2.5 Manage

Applying the Scrum framework was how I kept the whole project from derailing. I broke this massive, intimidating project down into manageable weekly sprints. During our check-ins, I would set clear goals for the week and honestly review what went wrong in the last one. This iterative management style kept me focused on the final working application instead of getting lost in technical rabbit holes. I also made sure to actively participate in bi-weekly team meetings, sharing my challenges and helping brainstorm solutions for other researchers' setups.

2.6 Advise

Working in an applied physics lab meant that the general understanding of software architecture and programming among the team was somewhat limited. At the beginning, I really struggled with this. I would catch myself giving highly technical software explanations that didn't make much sense to someone whose expertise was strictly in quantum optics. But as the project progressed, I actively worked on improving how I communicated technical programming concepts to people outside of my field. I learned to focus on how the code impacted the physical experiment rather than getting bogged down in software terms. Because I was able to bridge this communication gap, I ended up providing valuable feedback and solid recommendations for the future software development and maintenance of this setup.

2.7 Research

Coming from a computer science background, I had to adopt a seriously inquisitive mindset to tackle the physics side of this project. I dove into academic literature to understand how QIUP transfers image information without detecting the illuminating photons. When the standard manuals for the Thorlabs hardware failed me, I treated it as an applied re-

search problem. I systematically probed the undocumented APIs, logged the hardware's responses, and used logical deduction to map out the control functions. This hands-on, empirical approach was crucial for getting the piezoelectric stage to scan the interference fringes properly.

2.8 Professionalize

At the start of this internship, I honestly had barely any confidence. Stepping into an applied nanotechnology lab with a computer science background made the learning curve feel incredibly steep, especially when dealing with complex optical hardware and a lack of straightforward documentation. However, as the project progressed, I learned how to trust my foundational programming skills and combine them with the newly discovered physical concepts I have learned. By the end of the project, I was able to successfully bridge the gap between my prior software knowledge and the intricate physics of quantum imaging. It was a massive step forward for me to go from feeling completely out of my depth to ultimately engineering a robust, nicely working setup. This experience didn't just improve my technical versatility in Python and computer vision, it fundamentally built my professional confidence to tackle highly interdisciplinary challenges.

3 Self-analysis

Looking back at this project, I was thrown right into the deep end of Quantum Optics and Applied Nanotechnology. My main challenge was mastering the experimental data acquisition and complex signal analysis required for Quantum Imaging with Undetected Photons (QIUP).

Reflecting on my strengths, I demonstrated strong adaptability and resilience in software-hardware integration. I gained incredible hands-on experience orchestrating a highly synchronized setup. I successfully learned to integrate a CMOS camera and program a piezo-electric stage to scan interference fringes, diving deep into Thorlabs APIs to automate the hardware loop. On the software side, I leveraged my computer science background to process raw, phase-shifted image stacks using advanced Fourier transform techniques for phase retrieval.

However, this project also exposed several weaknesses. Initially, my lack of domain-specific physics knowledge significantly slowed down the design phase. I also noticed a tendency to become overly focused when troubleshooting complex hardware-software communication issues. I would often spend a significant amount of time trying to solve a coding problem independently, rather than recognizing when it would be more efficient to step back and consult the research team's expertise. Furthermore, my time-estimation skills were tested, as I consistently underestimated how long it would take to figure out how to properly work with and utilize the specific functions of undocumented SDKs.

Looking forward, I have identified clear opportunities for further self-development. The bottlenecks I encountered during image processing have inspired me to explore machine learning and computer vision as tools for real-time noise reduction and faster phase extraction. Additionally, deepening my knowledge of lower-level programming languages alongside Python will allow me to build more efficient, native hardware interfaces in future automation projects.

I truly poured my absolute best effort into mastering these physical and programmatic challenges. Pushing myself past my initial limits paid off immensely, and it is incredibly rewarding to see that dedication reflected in a final, robust system that completely exceeded my supervisor's expectations, while simultaneously providing me with a clear roadmap for my professional growth.

4 Personal development plan

During my studies I have developed an interest in vision control, image processing and object detection, and I would like to further develop my skills in these areas. I am also interested in learning more about machine learning and deep learning, and how they can be applied to image processing and computer vision. I would like to take courses on these topics and work on projects that involve them.

Another interest of mine is combining my software skill with hardware into advanced automation and control systems, in which i apply these vision control and image processing skills to create intelligent systems that can interact with the physical world. I would like to work on projects that involve building and programming robots, drones, cars or other autonomous systems.

To translate these interests into a concrete two-year outlook, I plan to pursue a Master's degree in Germany immediately following my bachelor's graduation. I am currently evaluating programs focused on either AI or Embedded Systems. A Master's in AI would allow me to dive deeper into machine learning architectures and deep learning frameworks, directly building on the computational imaging challenges I tackled during this internship. Alternatively, a program in Embedded Systems would deepen my knowledge in Applied Computer Science with advanced hardware control, empowering me to design the intelligent, automated setups I am passionate about.

Most importantly, I want to have an impact on the world and contribute to solving real-world problems. I want to use my skills and knowledge to create innovative solutions that can improve people's lives and make the world a better place. I would like to work on projects that have a positive social or environmental impact, and that can help address some of the biggest challenges facing humanity.